

```

sPython from keras.
optimizers import
Adam from keras.
layers import Dense,
Dropout, Activation,
Flatten, Input, LSTM
• Keras CNN layer-
sPython, from keras.
layers import Convo-
lution2D, MaxPooling2D

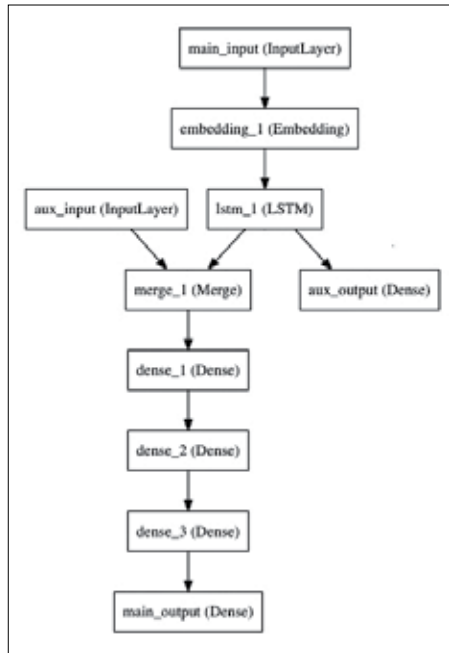
```

The next step involves adding of some utility functions to the Python environment by using the import function. Among the most important included are the ones for numpy utility, neural network architectures and hidden layers.

```

• from keras.utils
import np_utils,
from keras.datasets
import mnist

```



Keras Functional API

Running a sample program

Split the data into the required sets for training and testing, using the following code. Alternatively, you can choose to split the data using the `train_test_split` function or the `cv` syntax (cross validation)- (`X_train, y_train`), (`X_test, y_test`) = `testdata.load_data()`

Perform some necessary exploratory analysis of the data by using the matplotlib commands to compose graphs, distribution plots and correlation graphs for the inputs:-

```

• from matplotlib import pyplot as plt, plt.imshow(X_
train[0])

```

Users have to specify the dimensional depth of the input range by using the Theano backend. Theano is very useful for its potential to process large sums of visual media taking up gigabytes of space. It is also necessary to run and transform the dataset from one shape to another having a particular width, depth and height.